

Answer Keys to HW 8

$$1) \quad Y(\Omega) = \frac{-\frac{2}{3}e^{j\Omega}}{e^{j\Omega} + \frac{1}{2}} - \frac{\frac{2}{7}e^{j\Omega}}{e^{j\Omega} - \frac{1}{3}} + \frac{\frac{20}{21}e^{j\Omega}}{e^{j\Omega} - \frac{1}{10}}$$

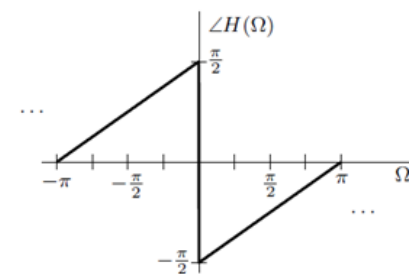
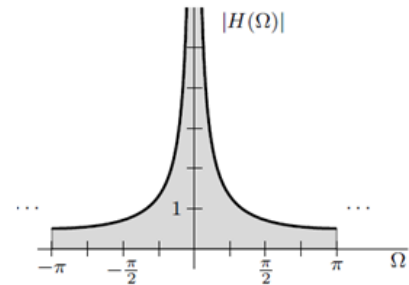
$$y[n] = \left(-\frac{2}{3} \left(-\frac{1}{2}\right)^n - \frac{2}{7} \left(\frac{1}{3}\right)^n + \frac{20}{21} \left(\frac{1}{10}\right)^n\right) u[n]$$

$$2) \quad H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = \frac{1}{1 - e^{-j\Omega}}$$

$$y_a[n] = 10 \cos(0.1n - 1.521)$$

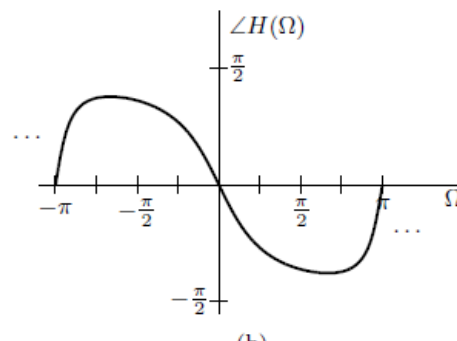
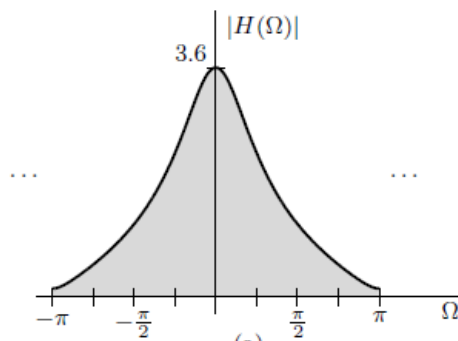
$$y_b[n] = 0.7071 \sin\left(\frac{\pi n}{2} - 0.7854\right)$$

$$y_c[n] = 0.52 \cos\left(\frac{4\pi n}{5} - 0.3142\right)$$

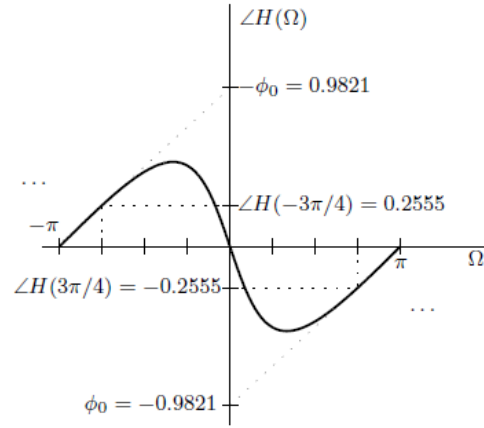
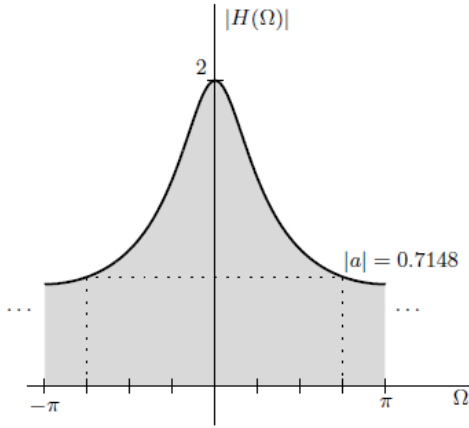


$$3) \quad H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = \frac{e^{j\Omega} + \frac{4}{5}}{e^{j\Omega} - \frac{1}{2}}$$

$$y[n] = 2.8590 \cos\left(0.5n - \frac{\pi}{3} - 0.6253\right) = 2.8590 \cos(0.5n - 1.6725)$$



$$4) \quad H(\Omega) = \frac{e^{j\Omega}}{e^{j\Omega} - \frac{1}{2}} = \frac{1}{1 - \frac{1}{2}e^{-j\Omega}}$$



$$H_{\text{glp}}(\Omega) = \begin{cases} |a|e^{j(\phi_0 - \Omega n_g)} & \frac{3\pi}{4} - 0.02\pi \leq \Omega \leq \frac{3\pi}{4} + 0.02\pi \\ |a|e^{j(-\phi_0 - \Omega n_g)} & -\frac{3\pi}{4} - 0.02\pi \leq \Omega \leq -\frac{3\pi}{4} + 0.02\pi \end{cases}$$

$$\Omega_c = \frac{3\pi}{4},$$

$$|a| = |H(3\pi/4)| = 0.7148,$$

$$n_g(\Omega) = -\frac{d}{d\Omega} \angle H(\Omega) = \frac{2 \cos(\Omega) - 1}{5 - 4 \cos(\Omega)}$$

$$n_g = \frac{2 \cos(3\pi/4) - 1}{5 - 4 \cos(3\pi/4)} = -0.3084,$$

$$\phi_0 = \angle H(3\pi/4) + \frac{3\pi}{4} n_g = -0.9821$$

$$y[n] = |a|v[n - n_g] \cos [\Omega_c(n - n_g) + \phi_0]$$

$$= 0.7148v[n + 0.3084] \cos \left[\frac{3\pi}{4}(n + 0.3084) - 0.9821 \right]$$

5)

$$X_c(\omega) = \begin{cases} TX(\omega T) & |\omega| \leq \pi/T \\ 0 & \text{otherwise} \end{cases}$$

$$= \frac{1}{40000} \left[1 - 4 \cos\left(\frac{\omega}{40000}\right) + 2 \cos\left(\frac{2\omega}{40000}\right) - 4 \cos\left(\frac{3\omega}{40000}\right) + 2 \cos\left(\frac{4\omega}{40000}\right) \right] \Pi\left(\frac{\omega}{80000\pi}\right)$$

6a) $x_{\downarrow}[n] = x[2n] = \frac{3}{16} \text{sinc}(3n/8).$

$M_{\max} = 5$ is the maximum downsampling factor.

b) the DTFT of $x_{\downarrow}[n] = \frac{3}{16} \text{sinc}(3n/8)$ is $X_{\downarrow}(\Omega) = \sum_{k=-\infty}^{\infty} \frac{1}{2} \Pi\left(\frac{\Omega - 2\pi k}{2(3\pi/8)}\right)$

$$X_{\downarrow}(\Omega) = \frac{1}{2} \sum_{k=-\infty}^{\infty} \left[\Pi\left(\frac{\Omega - 4\pi k}{2(3\pi/8)}\right) + \Pi\left(\frac{\Omega - 2\pi - 4\pi k}{2(3\pi/8)}\right) \right]$$

c) $Y_i(\Omega) = X(\Omega) = \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{2(3\pi/16)}\right)$

From Eq. (6.72), $Y_i(\Omega) = Y(M\Omega)H_i(\Omega),$

where $H_i(\Omega) = M \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{2\pi/M}\right)$

$$\begin{aligned} \text{Thus, } Y_i(\Omega) &= \left[\sum_{k=-\infty}^{\infty} \frac{1}{2} \Pi\left(\frac{2\Omega - 2\pi k}{2(3\pi/8)}\right) \right] \left[2 \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{\pi}\right) \right] \\ &= \left[\sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - \pi k}{3\pi/8}\right) \right] \left[\sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{\pi}\right) \right] \end{aligned}$$

7) a) $X_i(\Omega) = X(3\Omega)H_i(\Omega)$ $x_i[n] = \sum_{k=0}^{\infty} k\gamma^k \text{sinc}\left(\frac{n}{3} - k\right)$

$$= \frac{\gamma e^{j3\Omega}}{(e^{j3\Omega} - \gamma)^2} 3 \Pi\left(\frac{\Omega}{2\pi/3}\right)$$

b) The cascade of an ideal interpolator followed by a downsampler, both of factor L , is an identity system. Thus,

$$y_{\downarrow}[n] = x[n] = n\gamma^n u[n] \quad \text{and} \quad Y_{\downarrow}(\Omega) = X(\Omega) = \frac{\gamma e^{j\Omega}}{(\gamma - e^{j\Omega})^2}$$

